DISTRIBUTED TRACING / OBSERVABILITY

OBSERVABILITY IS NOT MONITORING

Monitoring tells you something is wrong. Observability tells you why. In a complex distributed system, dashboards show symptoms without revealing causes. Observability — correlated logs, metrics, and distributed traces — lets engineers ask arbitrary questions about system state they didn't know to ask before the incident.

KEY BENEFITS

- ✓ Reduces MTTR by 50%+ — engineers find root cause rather than hunting symptoms
- ✓ Enables debugging of novel failure modes with no pre-existing dashboard
- ✓ Eliminates hours of log grep and cross-team pinging during incidents
- ✓ Distributed traces reveal latency sources invisible to single-service monitoring

REAL-WORLD EXAMPLE

Honeycomb's research found teams with high observability resolve incidents 50% faster and experience 40% fewer repeat incidents — compounding into significantly less total downtime over time.

PRACTICAL EXAMPLE

Users report slow checkouts. The engineer opens a trace, sees 94% of latency in a call to the inventory service, correlates with a deployment 20 minutes ago, and rolls back the specific service in 7 minutes — without touching anything else.

REFLECTION QUESTION

During your last production incident, how long did it take to identify the root cause?